

# SYUMAIRA AZHAR

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Computer Science graduate with expertise in network infrastructure design, Linux administration, and full-stack web development. Proficient in PHP, JavaScript, Python, and MySQL, with a focus on optimizing system performance and troubleshooting complex issues. Seeking to leverage technical skills in Network Engineering, IT Infrastructure, or Systems Administration roles.

## PROFESSIONAL EXPERIENCE

### HQ Perbadanan Kemajuan Negeri Selangor (PKNS)

Oct 2024 – Present

#### IT Network Support / Network Support

Shah Alam, Selangor

- Optimized enterprise network infrastructure by designing and deploying VLAN segmentation, enhancing network security and scalability for 100+ users across HQ operations and branch offices.
- Architected high-availability connectivity protocols and implemented proactive network monitoring to maintain 99.9% uptime for critical business services and daily operations.
- Collaborated on multi-phase infrastructure upgrade projects, technical evaluation of vendor proposals, and drafting comprehensive tender documentation for high-level procurement.
- Performed initial troubleshooting for network connectivity and configuration incidents, escalating complex issues to higher support tiers when required.
- Modernized front-end responsiveness of the PKNS landing page using HTML, CSS, and Bootstrap, improving accessibility and user engagement for public-facing digital assets.

### AIM Connections Sdn Bhd

March 2021 – July 2021

#### IT Technician (Internship)

Damansara, Selangor

- Administered the Passenger Information System (PIS) web application, ensuring 24/7 service availability through automated security audits and routine software patching.
- Configured and deployed managed network switches to reinforce infrastructure reliability and throughput for real-time transportation data delivery.
- Mitigated critical system downtime in a live transportation environment through rapid on-site hardware maintenance and deep-dive technical troubleshooting.

## EDUCATION

### UNIVERSITI TEKNIKAL MALAYSIA MELAKA (UTEM), MELAKA, BACHELOR'S DEGREE IN COMPUTER SCIENCE (Computer Networking)

2021 – 2025

- CGPA: 3.49

### POLITEKNIK UNGKU OMAR (PUO), IPOH, DIPLOMA IN INFORMATION TECHNOLOGY

2018 – 2021

- CGPA: 3.90

## PROJECTS

### Enterprise Network Infrastructure Simulation

#### Academic Project (Network Engineering)

Architected a complex, segmented enterprise network environment to simulate production-grade communication and identity management.

- Designed a scalable enterprise architecture using Cisco routers and switches, implementing Inter-VLAN routing to optimize cross-departmental traffic flow and security segmentation.
- Deployed centralized network services including Active Directory for identity management, alongside DNS, DHCP, TFTP, and other services to mimic corporate infrastructure standards.
- Conducted rigorous network validation and performance testing within a virtualized environment to ensure high availability and protocol integrity.

## Job Tracking System and Portfolio System (Home-Lab)

*Personal Project (Infrastructure & Development)*

Developed and deployed a containerized CRUD application for career workflow automation, hosted on a custom-built home lab infrastructure.

- Built and managed a Proxmox VE home lab running Ubuntu Server for self-hosted services.
- Deployed a containerized full-stack application using Docker and Docker Compose.
- Developed a PHP and SQLite back-end for efficient data storage and retrieval.
- Published a self-hosted portfolio website using Cloudflare Tunnel, custom domain, DNS, and HTTPS for secure remote access.
- Developed Telegram API integrations to automate notifications for job applications, interview invitations, assessments, and upcoming deadlines.

## Smart Automated Gate System (IoT)

*Final Year Project - UTeM*

Designed and developed an IoT-based vehicle access control system with computer vision.

- Developed license plate recognition (LPR) pipelines using Python and OpenCV to automate real-time vehicle verification and access security.
- Architected a MySQL and PHP back-end to manage vehicle registration data and maintain comprehensive access logs for audit trails.
- Integrated Raspberry Pi hardware with Linux-based software stacks to create a functional end-to-end security solution for gated environments.

## Secure Portable Media Server (NAS)

*Final Year Project - PUO*

Built a secure Network Attached Storage (NAS) solution for encrypted file management and media delivery.

- Hardened Debian Linux by implementing user authentication, access controls, and secure Samba file sharing.
- Configured and optimized wireless infrastructure protocols to deliver low-latency media streaming across localized networks.
- Managed OS hardening, service deployment, troubleshooting, and availability testing.

## Bus Ticketing System

*Academic Project*

Developed a terminal-based booking system using Object-Oriented Programming (OOP) and MySQL.

- Designed relational MySQL database schemas supporting ACID-compliant transactions to maintain data integrity for ticket registration.
- Leveraged C++ Object-Oriented Programming (OOP) to build modular, maintainable software for real-time business workflow reporting.

## SKILLS

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**Network Engineering:** TCP/IP, VLAN, Routing, LAN, WAN, DNS, DHCP, Cisco IOS, Network Troubleshooting & Infrastructure Support

**Operating Systems & Virtualization:** Linux Administration, Windows Administration, Docker, Virtualization, Proxmox VE

**Databases & Back-end:** MySQL, PHP, Python, RESTful API, Relational Database Design, Samba

**Front-end Development:** HTML, CSS, JavaScript, Bootstrap, Web Development

**Development Tools:** Git, GitHub

**Identity Management:** Active Directory

**Networking Protocols:** Inter-VLAN Routing

**Language:** Malay and English

## RELEVANT CERTIFICATIONS

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|-------------------------------------------------------------------|-----------|
| <b>Huawei HCIA Datacomp Completion Courses</b> — Huawei           | Aug 2023  |
| <b>Coursera Designing for User Experience</b> — Microsoft         | June 2025 |
| <b>Coursera Crash Course on Python</b> — Google                   | June 2025 |
| <b>Coursera User Interface Design and Prototyping</b> — Microsoft | June 2025 |
| <b>Coursera Fundamentals of UI/UX Design</b> — Microsoft          | June 2025 |